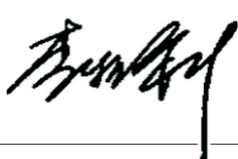


| 院教学指导委员会主任<br>(院长)                                                                | 学院分管教学<br>(副院长)                                                                   | 审核人<br>(专业责任教授团队负责人)                                                               | 执笔人 |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----|
|  |  |  | 张江  |

## 化工与制药类培养方案

### 一、大类所包含的专业

化工与制药类包含如下专业：化学工程与工艺、生物工程（产业计划）。

### 二、大类阶段课程设置

大类学科基础课程（必修）：高等数学、大学物理、无机化学、分析化学、工程制图、现代化工导论、生物工程导论。

大类学科选修课程（选修）：C语言程序设计基础、C语言程序设计基础实验、工程美学导论。

大类实践课程：军事训练、工程训练C。

## 化学工程与工艺专业培养方案

### Curriculum for Undergraduate of Chemical Engineering and Technology Major

#### 一、培养目标

本专业培养适应现代化工产业需求，德智体美劳全面发展的社会主义事业建设者和接班人。培养具有良好的人文科学素养和社会责任感，扎实的化学工程与工艺专业的基础理论和专业知识，具有创新意识与能力，拥有国际视野，能在化工及相关领域，特别是在煤化工、炭材料、能源化工等领域从事工程设计、科学研究、技术开发、运营管理等工作的高素质应用型人才。

本专业学生毕业五年左右应达到以下目标：

1、树立和践行社会主义核心价值观，具有良好的人文社会科学素养，强烈的社会责任感和职业道德，健康的身心，遵守工程伦理守则，促进社会可持续性发展；

2、具备化工产品、工艺过程和化工装备的设计开发能力，能熟练运用现代测试技术与相关工程软件从事产品开发和过程设计，能熟练系统运用数理及化工专业知识、工程技能和现代工具，遵守规范，并综合经济、社会、环境、安全等因素，解决化工及相关领域的复杂工程问题，具有较强的分析问题、设计方案的能力；

3、能够在化工及相关领域，特别是在煤化工、炭材料、能源化工等领域从事科学研究、工程设计、技术开发、运营管理工作，成为科学研究人员、技术骨干和管理者等，并取得相应专业资质和职称；

4、具有良好团队合作精神，沟通表达能力，组织协调能力，外语应用能力，适应团队工作环境，能与同事、专业客户和公众进行有效沟通；

5、具有自我发展和终身学习的意识和能力，具有国际视野和创新精神，能够主动适应职业环境的变化和发展，在化工及相关领域保持竞争力。

### **I. Training objectives**

This major cultivates builders and successors of socialist cause who meet the needs of modern chemical industry, with all-round development of morality, intelligence, sport, beauty and labor. Cultivating high-quality application-oriented talents with good humanities literacy and social responsibility, solid basic theory and professional knowledge of chemical engineering and technology, innovative consciousness and ability, and international vision, who can be engaged in engineering design, scientific research, technology development, operation and management in chemical industry and related fields, especially in coal chemical industry, carbon materials, energy chemical industry and other fields.

Students of this major should achieve the following goals about five years after graduation:

1. Establish and practice the socialist core values, have good humanities and social science literacy, strong sense of social responsibility and professional ethics, healthy body and mind, abide by the code of engineering ethics, and promote the sustainable development of society;

2. Have the design and development ability of chemical products, process and equipment, be able to skillfully use modern testing technology and related engineering software to engage in product development and process design, be able to skillfully use mathematical and chemical professional knowledge, engineering skills and modern tools systematically, comply with specifications, and integrate economic, social, environmental, safety and other factors, be able to solve complex engineering problems in chemical industry and related fields, and have strong ability to analyze problems and design schemes;

3. Have the ability to engage in scientific research, engineering design, technology development, operation and management, and other work in chemical industry and related fields, especially in coal chemical industry, carbon materials, energy chemical industry and other fields, become scientific researchers, technical backbones and managers, and obtain corresponding professional qualifications and titles;

4. Have good team spirit, communication skills, organization and coordination ability, foreign language application ability, can adapt to the team working environment, can effectively communicate with colleagues, professional customers and the public;

5. Have the awareness and ability of self-development and lifelong learning, international vision and innovative spirit, be able to actively adapt to the changes and development of professional environment, and maintain competitiveness in chemical industry and related fields.

### **二、毕业要求**

1、工程知识：能够将数学、自然科学、工程基础和专业知识用于解决化工领域的产品、装备和工艺过程设计与开发，以及生产中的复杂工程问题。

2、问题分析：能够应用数学、自然科学和工程科学的基本原理，识别、表达、并通过文献研究分析化工领域的复杂工程问题，以获得有效结论。

3、设计/开发解决方案：能够设计针对化工领域复杂工程问题的解决方案，设计满足特定化工过程要求的单元、系统和工艺流程，并能够在设计环节中体现创新意识，考虑社会、健康、安全、法律、文化以及环境等因素。

4、研究：能够根据化工领域复杂工程问题所涉及的科学原理，选择并运用适当的科学方法对复杂工程问题进行研究，包括设计实验、分析与解释数据，并通过信息综合得到合理有效的结论。

5、使用现代工具：能够针对化工领域复杂工程问题，开发、选择与使用恰当的技术、资源、现代工程工具和信息技术工具，包括对化工领域复杂工程问题的预测与模拟，并能够理解其局限性。

6、工程与社会：能够基于化工领域的工程背景知识，对工程项目进行合理分析，能针对性评价专业工程实践和复杂工程问题解决方案对社会、健康、安全、法律以及文化的影响，理解应承担的责任。

7、环境和可持续发展：建立环境和可持续发展的意识，能够理解和评价针对化工领域复杂工程问题的工程实践对环境、社会可持续发展的影响。

8、职业规范：具有人文社会科学素养、社会责任感，能够在工程实践中理解并遵守工程伦理守则和职业规范，履行责任。

9、个人和团队：能够在涵盖化工的多学科背景下的团队中承担个体、团队成员以及负责人的角色。

10、沟通：能够就化工领域复杂工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和 design 文稿、陈述发言、清晰表达或回应指令，并具备一定的国际视野，能够在跨文化背景下进行沟通和交流。

11、项目管理：理解并掌握化工领域工程管理原理与经济决策方法，并能在多学科环境中应用。

12、终身学习：具有自主学习和终身学习的意识，有不断学习和适应发展的能力。

## II. Requirements

1. Engineering knowledge: Be able to apply mathematics, natural science, engineering fundamentals and professional knowledge to the design and development of products, equipments and processes in chemical industry, as well as complex engineering problems in production.

2. Problem analysis: Be able to apply the basic principles of mathematics, natural science and engineering science to identify, express and analyze complex engineering problems in the chemical industry through literature research, so as to obtain effective conclusions.

3. Design/Develop solutions: Be able to design solutions to complex engineering problems in chemical industry, design units, systems and processes to meet specific chemical process requirements, and embody innovation awareness in design process, and consider social, health, safety, law, culture, environment and other factors.

4. Research: Be able to select and apply appropriate scientific methods to study complex engineering problems according to the scientific principles involved in complex engineering problems in the field of chemical engineering, including designing experiments, analyzing and interpreting data, and obtaining reasonable and effective conclusions through information integration.

5. Application of modern tools: Be able to develop, select and use appropriate technologies, resources, modern engineering tools and information technology tools for complex engineering problems in chemical industry, including prediction and simulation of complex engineering problems in chemical industry, and understand their limitations.

6. Engineering and society: Be able to conduct reasonable analysis of engineering projects based on engineering background knowledge in chemical industry, evaluate the impact of professional engineering practices and complex engineering problem solutions on society, health, safety, law and culture, and understand the responsibilities to be undertaken.

7. Environment and sustainable development: To develop awareness of environment and sustainable

development, and to understand and evaluate the impact of engineering practices on environmental and social sustainable development of complex engineering problems in the field of chemical engineering.

8. Professional norms: With humanistic and social science literacy and social responsibility, can understand and abide by the code of engineering ethics and professional norms in engineering practice and fulfill responsibilities.

9. Individual and teamwork: Be able to take on individual, team member and leader roles in a multidisciplinary team covering chemical engineering.

10. Communication: Can communicate effectively with peers and the public on complex engineering problems in chemical industry, including writing reports and design documents, making presentations, clearly expressing or responding to instructions, and have a certain international vision, and be able to communicate and exchange in a cross-cultural context.

11. Project management: Understand and master engineering management principles and economic decision-making methods in chemical industry, and can apply them in a multidisciplinary environment.

12. Lifelong learning: Have the consciousness of self-learning and lifelong learning, have the ability of continuous learning and adapting to development.

附：培养目标实现矩阵

|         | 培养目标 1 | 培养目标 2 | 培养目标 3 | 培养目标 4 | 培养目标 5 |
|---------|--------|--------|--------|--------|--------|
| 毕业要求 1  |        | √      |        |        |        |
| 毕业要求 2  |        | √      |        |        |        |
| 毕业要求 3  |        | √      |        |        |        |
| 毕业要求 4  |        | √      | √      |        |        |
| 毕业要求 5  |        | √      | √      |        |        |
| 毕业要求 6  | √      |        | √      |        |        |
| 毕业要求 7  | √      |        | √      |        |        |
| 毕业要求 8  | √      |        |        | √      |        |
| 毕业要求 9  |        |        |        | √      |        |
| 毕业要求 10 |        |        |        | √      | √      |
| 毕业要求 11 |        |        |        | √      |        |
| 毕业要求 12 |        |        |        |        | √      |

### 三、专业主干课程

有机化学、物理化学、化工热力学、化工原理、化工过程分析与合成、化工环保与安全技术、化工工程设计与技术经济分析、化工设备与材料、现代化工导论、化学反应工程、能源化学。

### III. Core courses

Organic Chemistry, Physical Chemistry, Chemical Engineering Thermodynamics, Principles of Chemical Engineering, Chemical process analysis and integration, Environmental Protection and Safety Technologies of Chemical, Chemical Engineering Design and Techno-Economics Analysis, Chemical Engineering Equipment and Materials, Introduction to Modern Chemical Engineering, Chemical Reaction Engineering, Energy Chemistry.

#### 四、基本学制：四年

#### IV. Recommended length of the program: 4 years

#### 五、授予学位：工学学士

#### V. Degree: Bachelor of Engineering

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予工学学士学位。

#### 六、毕业学分要求：160学分

| 课程类型       |    | 学分要求 | 课程类型     |    | 学分要求 |
|------------|----|------|----------|----|------|
| 1、公共课程平台   |    | 45   | 3、专业课程模块 |    | 46   |
| 公共基础课程     |    | 29   | 专业必修课程   |    | 39.5 |
| 通识教育课程     | 必修 | 9    | 专业选修课程   |    | 6.5  |
|            | 选修 | 7    | 4、实践教学模块 |    | 18   |
| 2、学科基础平台   |    | 45   | 大类实践课程   |    | 3    |
| 大类专业学科基础课程 | 必修 | 25.5 | 专业实践课程   | 必修 | 15   |
|            | 选修 | 0    |          |    |      |
| 专业学科基础课程   |    | 19.5 | 5、素质拓展模块 |    | 6    |

\* 通识教育选修课7学分包括：理工类、医学类、人文社科类、经济管理类中选择3学分；思想政治及新时代素质教育类选择2学分；美育教育类选择2学分。

#### VI. Credits required for graduation: 160 credits

| Type of courses                |                  | Academic credits | Type of courses                    |                  | Academic credits |
|--------------------------------|------------------|------------------|------------------------------------|------------------|------------------|
| 1.Common Courses               |                  | 45               | 3. Specialized Courses             |                  | 46               |
| Common Basic Courses           |                  | 29               | Required Courses                   |                  | 39.5             |
| General Education Courses      | Required Courses | 9                | Elective Courses                   |                  | 6.5              |
|                                | Elective Courses | 7                | 4.Practicum and Internship Courses |                  | 18               |
| 2.General Disciplinary Courses |                  | 45               | Large Category Practical Courses   |                  | 3                |
| Large Category Basic Courses   | Required Courses | 25.5             | Disciplinary Practical Courses     | Required Courses | 15               |
|                                | Elective Courses | 0                |                                    |                  |                  |
| Disciplinary Basic Courses     | Required Courses | 19.5             | 5. Quality Development Courses     |                  | 6                |

## 七、学分比例

### VII. Ratio of Credits

#### 1、必修选修学分比例

The proportion of compulsory elective credits

| 类别 | 学分    | 占总学分比例 |
|----|-------|--------|
| 必修 | 146.5 | 91.6%  |
| 选修 | 13.5  | 8.4%   |

#### 2、实践教学环节学分比例

The Proportion of credits in practice teaching

| 类别     | 学分     | 占总学分比例 |
|--------|--------|--------|
| 实践教学环节 | 实验教学学分 | 24.4   |
|        | 实践教学模块 | 18     |
|        | 素质拓展模块 | 6      |
|        |        | 30.2%  |

## 八、辅修

修读本专业辅修课程达到26学分且主修专业达到毕业要求者，颁发辅修专业证书。修读本专业辅修课程达到40学分（含实践教学），并达到辅修学位授予条件的，颁发辅修学位证书。

### VIII. Minor course

The minor certificate will be issued to those who have completed 26 credits of minor courses and their own major has met the graduation requirements. The minor degree certificate will be issued to those who have completed 40 credits of minor courses (including practical teaching) and meet the conditions for awarding minor degree.

## 九、毕业要求实现矩阵

## IX.Graduation Realization Matrix

| 课程名称                 | 化学工程与工艺专业毕业要求 |     |     |     |     |     |     |     |     |      |      |      |
|----------------------|---------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
|                      | (1)           | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| 思想道德与法治              |               |     |     |     |     | √   |     | √   |     |      |      |      |
| 中国近现代史纲要             |               |     |     |     |     |     |     | √   |     |      |      |      |
| 马克思主义基本原理            |               |     |     |     |     |     |     | √   |     |      |      |      |
| 毛泽东思想和中国特色社会主义理论体系概论 |               |     |     |     |     |     | √   | √   |     |      |      |      |
| 形势与政策                |               |     |     |     |     | √   | √   | √   |     | √    |      | √    |
| 军事理论                 |               |     |     |     |     |     |     |     | √   |      |      |      |
| 军事训练                 |               |     |     |     |     |     |     |     | √   |      |      |      |
| 体育                   |               |     |     |     |     |     |     |     | √   |      |      |      |
| 公益劳动                 |               |     |     |     |     |     |     | √   | √   |      |      |      |
| 大学生心理健康教育            |               |     |     |     |     |     |     | √   |     | √    |      |      |
| 职业生涯规划与就业指导          |               |     |     |     |     | √   |     | √   |     |      | √    | √    |
| 创业学基础                |               |     |     |     |     |     |     |     |     |      | √    | √    |
| 大学计算机基础              |               |     |     |     | √   |     |     |     |     |      |      |      |
| 英语                   |               |     |     |     |     |     |     |     |     | √    |      |      |
| 高等数学                 | √             |     |     |     |     |     |     |     |     |      |      |      |
| 线性代数                 | √             | √   |     |     |     |     |     |     |     |      |      |      |
| 数学建模                 | √             | √   |     |     |     |     |     |     |     |      |      |      |
| 概率论及数理统计             | √             | √   |     |     |     |     |     |     |     |      |      |      |
| 工程制图 B               | √             |     | √   |     | √   |     |     |     |     |      |      |      |
| 无机化学                 | √             |     |     |     |     |     |     |     |     |      |      |      |
| 无机化学实验               |               |     |     | √   |     |     |     |     |     |      |      |      |
| 分析化学                 |               |     |     |     | √   |     |     |     |     |      |      |      |
| 分析化学实验               |               |     |     | √   |     |     |     |     |     |      |      |      |
| 现代化工导论               |               |     |     |     |     |     | √   |     |     | √    |      |      |
| 生物工程导论               |               |     |     |     |     |     | √   |     |     | √    |      |      |
| 大学物理                 | √             |     |     |     |     |     |     |     |     |      |      |      |
| 大学物理实验               |               |     |     | √   |     |     |     |     |     |      |      |      |
| 电路电子技术               |               |     |     | √   | √   |     |     |     |     |      |      |      |

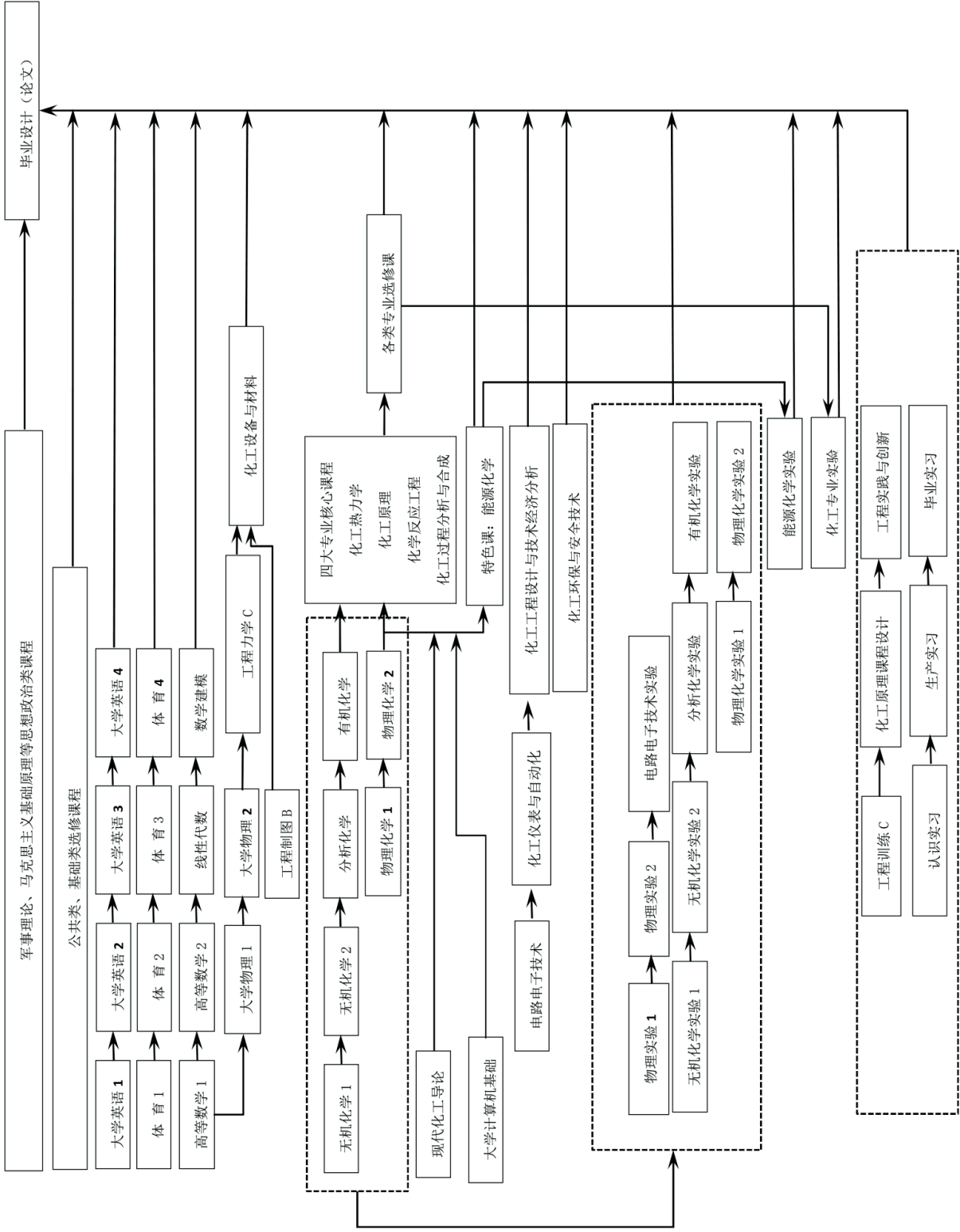
| 课程名称          | 化学工程与工艺专业毕业要求 |     |     |     |     |     |     |     |     |      |      |      |
|---------------|---------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
|               | (1)           | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| 电路电子技术实验      |               |     |     | √   | √   |     |     |     |     |      |      |      |
| 工程力学 C        | √             | √   | √   |     |     |     |     |     |     |      |      |      |
| 化工仪表与自动化      | √             |     |     | √   | √   |     |     |     |     |      |      |      |
| 计算机辅助化工制图与识图  | √             |     |     |     | √   |     |     |     |     |      |      |      |
| C 语言程序设计基础    |               |     |     |     | √   |     |     |     |     |      |      |      |
| C 语言程序设计基础实验  |               |     |     |     | √   |     |     |     |     |      |      |      |
| 传递过程基础        | √             | √   | √   |     |     |     |     |     |     |      |      |      |
| 实验设计与数据处理     |               | √   |     | √   |     |     |     |     |     |      |      |      |
| 化工工程放大原理      | √             | √   | √   |     |     |     |     |     |     |      |      |      |
| 高分子物理与化学      | √             |     |     |     |     |     |     |     |     |      |      |      |
| 分离工程          | √             | √   | √   |     |     |     |     |     |     |      |      |      |
| 专业英语          |               |     |     |     |     |     |     |     |     | √    |      |      |
| 现代分析与测试技术     |               |     |     | √   | √   |     |     |     |     |      |      |      |
| 化工热力学 B       | √             | √   |     |     |     |     |     |     |     |      |      |      |
| 物理化学          | √             | √   |     |     |     |     |     |     |     |      |      |      |
| 物理化学实验        |               | √   |     | √   |     |     |     |     |     |      |      |      |
| 化工设备与材料 A     | √             | √   | √   |     |     |     |     |     |     |      |      |      |
| 化工工程设计与技术经济分析 |               | √   | √   |     |     | √   |     |     |     |      | √    |      |
| 化工过程分析与合成 B   | √             | √   | √   | √   | √   |     |     |     |     |      |      |      |
| 化学反应工程 A      | √             | √   |     |     | √   |     |     |     |     | √    |      |      |
| 化学反应工程实验      | √             | √   |     | √   |     |     |     |     |     |      |      |      |
| 有机化学          | √             |     | √   |     |     |     |     |     |     |      |      |      |
| 有机化学实验        |               |     |     | √   |     |     |     |     |     |      |      |      |
| 化工原理          | √             | √   |     | √   |     |     |     |     |     |      |      |      |
| 化工原理实验        | √             | √   |     | √   |     |     |     |     |     |      |      |      |
| 能源化学          | √             | √   |     |     |     |     | √   |     |     | √    |      |      |
| 能源化学实验        |               |     |     | √   |     |     |     |     | √   |      |      |      |
| 化工环保与安全技术     |               | √   | √   |     |     | √   | √   | √   |     |      |      |      |
| 化工专业实验        |               | √   | √   | √   |     |     |     |     |     | √    |      |      |
| 燃气工程          | √             | √   | √   |     |     |     | √   |     |     |      |      |      |
| 石油化学与石油炼制工程   | √             |     | √   |     |     |     | √   |     |     |      |      |      |
| 精细有机合成及工艺学    |               | √   | √   | √   |     |     | √   |     |     |      |      |      |

| 课程名称         | 化学工程与工艺专业毕业要求 |     |     |     |     |     |     |     |     |      |      |      |
|--------------|---------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
|              | (1)           | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| 工业药剂学        |               | √   | √   | √   |     |     |     |     |     |      |      |      |
| 炭材料科学与工程基础 A |               | √   |     | √   |     |     |     |     |     |      |      |      |
| 药品生产质量管理工程   |               |     | √   |     |     | √   |     | √   |     |      |      |      |
| 化学制药工艺学      |               | √   | √   | √   |     |     |     |     |     |      |      |      |
| 化工工艺学        |               |     | √   |     |     |     |     |     |     |      | √    |      |
| 能源化工工艺学      |               |     | √   |     |     |     |     |     |     |      | √    |      |
| 化工过程计算机模拟    | √             |     | √   | √   | √   |     |     |     |     |      |      |      |
| 化学产品设计基础     |               | √   | √   |     |     |     |     |     |     |      |      |      |
| 化工仿真实验       |               | √   |     | √   | √   |     |     |     |     |      |      |      |
| 应用催化         | √             |     |     |     | √   |     |     |     |     |      |      |      |
| 燃料燃烧         | √             | √   |     |     |     | √   | √   |     |     |      |      |      |
| 药物化学与新药研发    |               | √   | √   | √   | √   |     |     |     |     |      |      |      |
| 纳米材料基础       | √             | √   |     |     |     |     |     |     |     |      |      |      |
| 工程美学导论       | √             |     | √   |     |     | √   |     | √   |     |      |      |      |
| 高分子材料加工工艺学   |               | √   |     | √   |     | √   |     |     |     |      |      |      |
| 绿色化学         | √             | √   | √   |     |     | √   |     |     |     |      |      |      |
| 材料化学         | √             |     |     |     | √   |     |     |     |     |      |      |      |
| 化学开放实验       | √             |     | √   | √   | √   |     |     |     |     |      |      |      |
| 应用电化学        | √             |     |     |     |     |     |     |     |     |      |      |      |
| 工程训练 C       |               |     | √   |     |     | √   |     |     | √   |      |      |      |
| 工程实践与创新      |               |     | √   |     | √   |     |     |     | √   | √    | √    |      |
| 化工原理课程设计     |               | √   | √   |     | √   |     |     |     | √   |      |      |      |
| 认识实习         |               |     |     |     | √   | √   |     | √   |     |      |      |      |
| 生产实习         |               |     |     |     |     | √   | √   | √   | √   | √    |      |      |
| 毕业实习         |               |     |     | √   | √   | √   |     |     |     |      | √    |      |
| 毕业设计(论文)     |               |     | √   |     | √   | √   |     |     |     | √    | √    | √    |
| 创新创业         |               |     |     |     |     |     |     |     | √   |      | √    | √    |
| 第二课堂         |               |     |     |     |     |     |     |     | √   | √    |      |      |

(四年制本科 2021级 化学工程与工艺专业)

十、课程修读进程表

X.Course review Process Map



# 生物工程专业“战略性新兴产业（支柱）产业计划”培养方案

## Curriculum for Undergraduate of Biological Engineering Major “Talents of Strategic and Emerging (Pillar) Industry”

### 一、培养目标

本专业根据学校“立足湖北、面向全国”的办学定位及人才培养目标，结合生物工程专业人才需求和国家经济发展需要，旨在面向全国医药、食品、环保、诊断等生物工程领域，培养德、智、体、美、劳全面发展的社会主义事业合格建设者和可靠接班人。培养具有强烈社会责任感、良好的道德修养和健康的心理素质、理论基础扎实、综合素质高、实践能力强，具备一定的国际视野，具有创新能力和持续学习能力，掌握生物工程专业基本原理、技能、研究方法和管理知识，能够解决生物加工过程中复杂工程问题，可以在以生物工程为基础的相关领域从事生产、设计、管理和新技术研究、新产品开发等工作、适应社会发展需要的高素质工程技术创新型人才。

学生在毕业5年左右，应达到如下目标：

- 1、具有良好的人文科学素养、社会责任感和工程职业道德修养，具备扎实的工程实践能力，能够根据国家和产业发展需求，设计、实施符合法律政策规定、有益于经济、环境和社会发展的生物工程相关项目，并能与业界及社会大众进行有效的沟通交流；
- 2、能够适应独立和团队工作环境，运用所学理论知识与实践技能，在生物工程产业界或学术界发现、分析并解决复杂的生物工程问题，展示个人的职业能力和价值；
- 3、能够与时俱进，具备对生物工程领域新理论、新产品、新工艺、新技术和新装备进行研究、开发的实践能力与创新能力。；
- 4、具有国际视野和良好的外语应用能力，能够在不同职能团队中发挥关键作用，并具备承担领导角色的能力；
- 5、能够通过继续教育或其它途径进行终身学习，适应职业发展，在生物工程领域保持职业竞争力。

### I.Training objectives

According to the school running orientation and talent training goal of "based on Hubei and facing the whole country", combined with the talent demand of bioengineering specialty and the needs of national economic development, this major aims to cultivate qualified builders and reliable successors of socialist cause with all-round development of morality, intelligence, sports, beauty and labor in the bioengineering fields of medicine, food, environmental protection and diagnosis. To cultivate students with strong sense of social responsibility, good moral cultivation and healthy psychological quality, solid theoretical foundation, high comprehensive quality, strong practical ability, certain international vision, innovation ability and continuous learning ability, master the basic principles, skills, research methods and management knowledge of bioengineering, and be able to solve complex engineering problems in the process of biological processing, It can be engaged in production, design, management, new technology research and new product development in related fields based on bioengineering and meet the needs of social development.

Students should achieve the following goals about 5 years after graduation:

1. Have good humanities literacy, social responsibility and engineering professional ethics, have solid

engineering practice ability, be able to design and implement bioengineering related projects in line with laws and policies and beneficial to economic, environmental and social development according to the needs of national and industrial development, and be able to effectively communicate with the industry and the public;

2. Be able to adapt to the independent and team work environment, use the theoretical knowledge and practical skills, discover, analyze and solve complex bioengineering problems in the bioengineering industry or academia, and show personal professional ability and value;

3. Be able to keep pace with the times, and have the practical and innovative ability to research and develop new theories, new products, new processes, new technologies and new equipment in the field of bioengineering.;

4. Have international vision and good foreign language application ability, be able to play a key role in different functional teams, and have the ability to assume a leading role;

5. Be able to carry out lifelong learning through continuing education or other ways, adapt to career development, and maintain professional competitiveness in the field of bioengineering.

## 二、毕业要求

本专业培养的学生在毕业时，通过本科阶段的培养和训练，能够获得下列知识、能力和素养：

1、工程知识：掌握解决生物加工过程问题的数学、工程基础、自然科学和专业基础知识；能够将数学、自然科学、工程基础和生物工程专业知识用于解决生物工程领域复杂工程问题。

2、问题分析：能够应用数学、自然科学和工程科学的基本原理同时结合文献研究，识别、表达和分析分析生物工程专业领域复杂工程问题，以获得有效结论。

3、设计/开发解决方案：能够设计生物工程领域复杂工程问题的解决方案，设计满足特定需求的工艺流程及技术、单元（部件）、以及系统，并能够在设计环节中体现创新意识，充分考虑健康、安全、环境、法律、以及社会与文化等因素。

4、研究：能够基于科学原理并采用科学方法对生物工程领域复杂工程问题进行研究，包括实验设计、获取与分析数据，并通过信息综合得到合理有效的结论。

5、使用现代工具：能够针对生物工程复杂工程问题，开发、选择与使用恰当的现代技术、资源、科学仪器、工程工具和信息技术工具，能对生物工程复杂工程问题进行模拟、分析和预测，并能够理解其局限性。

6、工程与社会：能够基于工程相关背景知识进行合理分析，评价生物工程实践和复杂工程问题解决对社会、健康、安全、法律以及文化的影响，并理解应承担的责任。

7、环境和可持续发展：能够理解和评价针对生物工程领域复杂工程问题的工程实践对环境、社会可持续发展的影响。

8、职业规范：具有人文社会科学素养、社会责任感，能够在生物工程实践中理解并遵守工程职业道德和规范，履行责任。

9、个人和团队：能够在多学科背景下的团队中承担个体、团队成员以及负责人的角色。

10、沟通：能够就生物工程领域问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计文稿、陈述发言、清晰表达或回应指令，并具备一定的国际视野，能够在跨文化背景下进行沟通和交流。

11、项目管理：理解并掌握生物工程领域工程管理原理与经济决策方法，并能在多学科环境中应用。

12、终身学习：具有自主学习和终身学习的意识，有不断学习和适应发展的能力。

## II. Requirements

1. Engineering knowledge: Master the mathematics, engineering foundation, natural science and professional

knowledge to solve the problems of biological processing process; Have the ability to use mathematics, natural science, engineering foundation and bioengineering professional knowledge to solve complex engineering problems in bioengineering field.

2. Problem analysis: Have the ability to apply the basic principles of mathematics, natural science and Engineering Science, and combine with literature research to identify, express and analyze complex engineering problems in the field of bioengineering, so as to obtain effective conclusions.

3. Design or develop solutions: Have the ability to design solutions to complex engineering problems in the field of bioengineering, design process flow and technology, units (components), and systems to meet specific needs, embody innovative consciousness in the design process, and fully consider factors such as health, safety, environment, law, society and culture.

4. Experimental design and information processing: Have the ability to study complex engineering problems in the field of bioengineering based on scientific principles and scientific methods, including experimental design, data acquisition and analysis, and get reasonable and effective conclusions through information synthesis.

5. Application of modern tools: Have the ability to develop, select and use appropriate modern technologies, resources, scientific instruments, engineering tools and information technology tools for complex bioengineering problems, simulate, analyze and predict complex bioengineering problems, and understand their limitations.

6. The sense of social responsibility for engineers: Have the ability to conduct reasonable analysis based on engineering related background knowledge, evaluate the impact of bioengineering practice and complex engineering problem solutions on society, health, safety, law and culture, and understand the responsibilities.

7. Environment and sustainable development: Have the ability to understand and evaluate the impact of engineering practice aiming at complex engineering problems in the field of bioengineering on environmental and social sustainable development.

8. Professional ethics and criteria: Have humanities and social science literacy, sense of social responsibility, be able to understand and abide by engineering professional ethics and norms in bioengineering practice, and fulfill their responsibilities.

9. Teamwork: Have the ability to play the role of individual, team member and leader in a multidisciplinary team.

10. Communication: Have the ability to effectively communicate with industry peers and the public on bioengineering issues, including writing reports and design papers, presenting statements, clearly expressing or responding to instructions, and having a certain international perspective and being able to communicate and communicate in cross-cultural context.

11. Project management: Understand and master the principles of engineering management and economic decision-making methods in the field of bioengineering, and be able to apply them in a multidisciplinary environment.

12. Lifelong learning: Have the consciousness of self-learning and lifelong learning, have the ability of continuous learning and adapting to development.

附：培养目标实现矩阵

|         | 培养目标 1 | 培养目标 2 | 培养目标 3 | 培养目标 4 | 培养目标 5 |
|---------|--------|--------|--------|--------|--------|
| 毕业要求 1  | √      | √      | √      |        |        |
| 毕业要求 2  |        | √      | √      |        |        |
| 毕业要求 3  | √      | √      | √      |        | √      |
| 毕业要求 4  |        | √      | √      |        |        |
| 毕业要求 5  |        | √      | √      | √      | √      |
| 毕业要求 6  | √      |        | √      |        | √      |
| 毕业要求 7  | √      | √      | √      |        |        |
| 毕业要求 8  | √      |        |        |        |        |
| 毕业要求 9  |        | √      |        | √      |        |
| 毕业要求 10 |        | √      |        | √      |        |
| 毕业要求 11 | √      |        | √      | √      |        |
| 毕业要求 12 |        |        |        |        | √      |

### 三、专业主干课程

有机化学、生物化学、微生物学、化工原理、基因工程、发酵工程、生物分离工程、生化反应工程原理、生物工程工厂设计概论。

### III. Core courses

Organic chemistry, Biochemistry, Microbiology, Chemical engineering principles, genetic engineering, Fermentation engineering, Bio-separation engineering, Principles of Biochemical reaction Engineering, Introduction to Bioengineering Plant Design.

### 四、基本学制：四年

### IV. Recommended length of the program: 4 years

Generally, students need 4 years to complete their studies. It is also acceptable to complete all required credits in flexible 3 to 6 years.

### 五、授予学位：工学学士

### V. Degree: Bachelor of Engineering

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予工学学士学位。

## 六、毕业学分要求：160学分

| 课程类型     |    | 学分要求         | 课程类型     |    | 学分要求 |
|----------|----|--------------|----------|----|------|
| 1、公共课程平台 |    | 45           | 3、专业课程模块 |    | 33   |
| 公共基础课程   |    | 29           | 专业必修课程   |    | 21   |
| 通识教育课程   | 必修 | 9            | 专业选修课程   |    | 12   |
|          | 选修 | 7            | 4、实践教学模块 |    | 21   |
| 2、学科基础平台 |    | 55           | 大类实践课程   |    | 3    |
| 大类学科基础课程 | 必修 | 25.5         | 专业实践课程   | 必修 | 18   |
|          | 选修 | 0            |          |    |      |
| 专业学科基础课程 |    | 必修 24+选修 5.5 | 5、素质拓展模块 |    | 6    |

\* 通识教育选修课7学分包括：理工类、医学类、人文社科类、经济管理类中选择3学分；思想政治及新时代素质教育类选择2学分；美育教育类选择2学分。

## VI. Credits required for graduation; 160 credits

| Type of courses                |                  | Academic credits | Type of courses                    |                  | Academic credits |
|--------------------------------|------------------|------------------|------------------------------------|------------------|------------------|
| 1.Common Courses               |                  | 45               | 3. Specialized Courses             |                  | 33               |
| Common Basic Courses           |                  | 29               | Required Courses                   |                  | 21               |
| General Education Courses      | Required Courses | 9                | Elective Courses                   |                  | 12               |
|                                | Elective Courses | 7                | 4.Practicum and Internship Courses |                  | 21               |
| 2.General Disciplinary Courses |                  | 55               | Large Category Practical Courses   |                  | 3                |
| Large Category Basic Courses   | Required Courses | 25.5             | Disciplinary Practical Courses     | Required Courses | 18               |
|                                | Elective Courses | 0                |                                    |                  |                  |
| Disciplinary Basic Courses     |                  | 29.5             | 6. Quality Development Courses     |                  | 6                |

## 七、学分比例

### VII. Ratio of Credits

#### 1、必修选修学分比例

The proportion of compulsory elective credits

| 类别 | 学分    | 占总学分比例 |
|----|-------|--------|
| 必修 | 135.5 | 84.7%  |
| 选修 | 24.5  | 15.3%  |

#### 2、实践教学环节学分比例

The Proportion of credits in practice teaching

| 类别     | 学分     | 占总学分比例 |
|--------|--------|--------|
| 实践教学环节 | 实验教学学分 | 26.3   |
|        | 实践教学模块 | 22     |
|        | 素质拓展模块 | 6      |
|        |        | 33.9%  |

## 八、辅修

修读本专业辅修课程达到26学分且主修专业达到毕业要求者，颁发辅修专业证书。修读本专业辅修课程达到40学分（含实践教学），并达到辅修学位授予条件的，颁发辅修学位证书。

### VIII. Minor course

The minor certificate will be issued to those who have completed 26 credits of minor courses and their own major has met the graduation requirements. The minor degree certificate will be issued to those who have completed 40 credits of minor courses (including practical teaching) and meet the conditions for awarding minor degree.

## 九、毕业要求实现矩阵

## IX. Graduation Realization Matrix

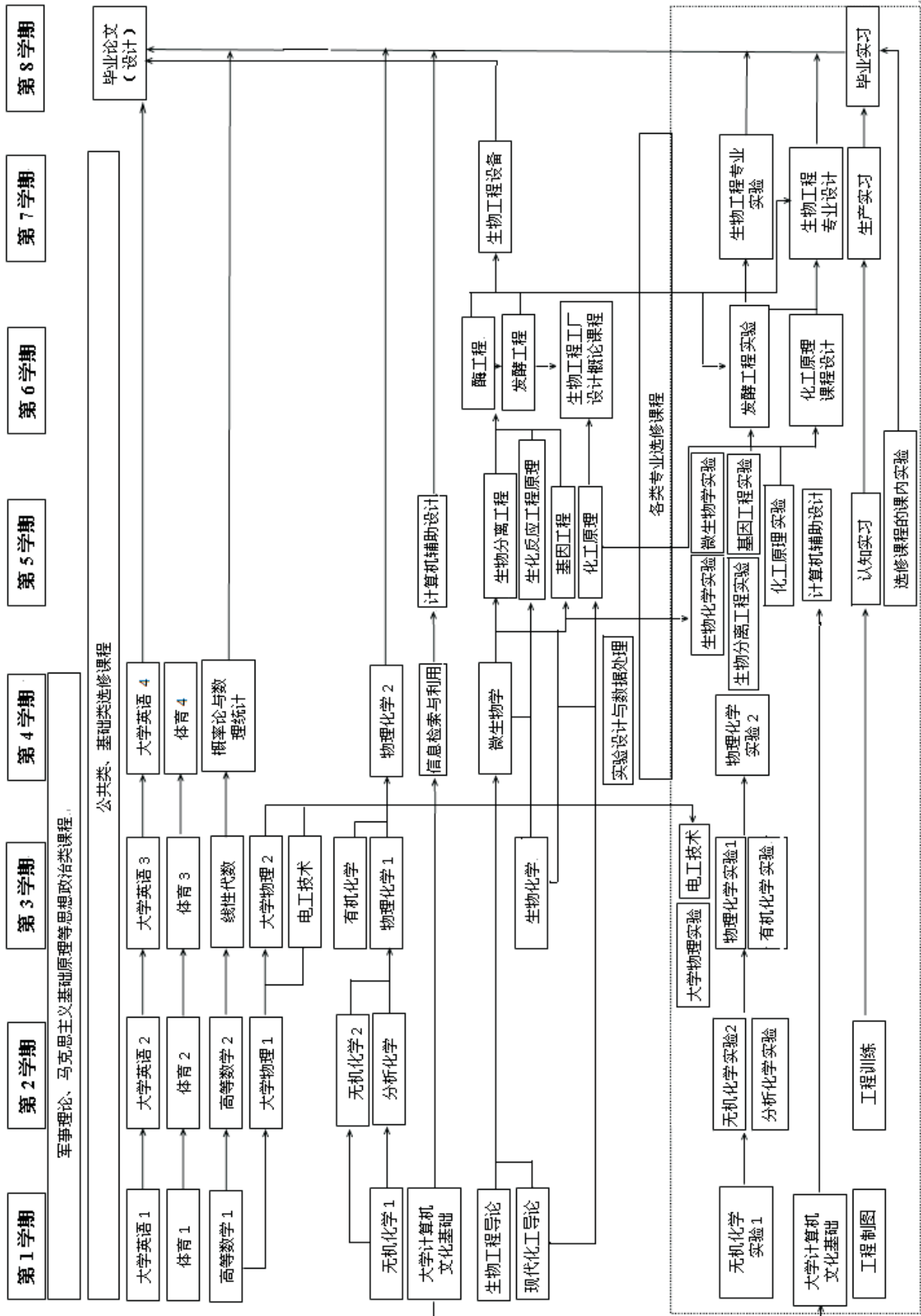
| 课程名称                 | 生物工程专业“产业计划”毕业要求 |     |     |     |     |     |     |     |     |      |      |      |
|----------------------|------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
|                      | (1)              | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| 思想道德与法治              |                  |     |     |     |     | √   | √   | √   | √   |      |      | √    |
| 中国近现代史纲要             |                  |     |     |     |     |     |     | √   |     |      |      |      |
| 马克思主义基本原理            |                  |     |     |     |     |     |     | √   |     |      |      |      |
| 毛泽东思想和中国特色社会主义理论体系概论 |                  |     |     |     |     |     |     | √   |     |      |      |      |
| 形势与政策                |                  |     |     |     |     |     |     | √   |     |      |      | √    |
| 军事课                  |                  |     |     |     |     |     |     |     | √   |      |      |      |
| 大学计算机文化基础 A          | √                |     |     |     | √   |     |     |     |     |      |      |      |
| 体育                   |                  |     |     |     |     |     |     | √   | √   |      |      |      |
| 大学综合英语               | √                |     |     |     |     |     |     |     |     |      |      | √    |
| 大学生心理健康教育            |                  |     |     |     |     | √   |     | √   | √   |      |      | √    |
| 公益劳动                 |                  |     |     |     |     | √   |     |     |     |      |      |      |
| 职业生涯规划与就业指导          |                  |     |     |     |     | √   | √   | √   | √   |      | √    | √    |
| 高等数学 B               | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 线性代数                 | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 概率论及数理统计 B           | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 大学物理 B               | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 大学物理实验 B             | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 计算机程序设计基础 C          | √                |     |     |     | √   |     |     |     |     |      |      |      |
| 工程制图 B               | √                | √   |     |     | √   |     |     |     |     |      |      |      |
| 生物化学                 | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 生物化学实验               | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 生物工程导论               | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 现代化工导论               | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 分析化学 B               | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 分析化学实验 B             | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 无机化学 A               | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 无机化学实验 A             | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 有机化学 B               | √                | √   |     |     |     |     |     |     |     |      |      |      |

| 课程名称        | 生物工程专业“产业计划”毕业要求 |     |     |     |     |     |     |     |     |      |      |      |
|-------------|------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
|             | (1)              | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| 有机化学实验 B    | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 物理化学 B      | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 物理化学实验 B    | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 电工技术        |                  |     |     |     | √   | √   |     |     |     |      |      |      |
| 管理学原理       |                  |     |     |     |     | √   |     | √   |     |      | √    |      |
| 信息检索与利用     |                  |     |     |     | √   |     |     |     |     |      |      | √    |
| 计算机辅助设计 CAD | √                |     |     |     | √   |     |     |     |     |      |      |      |
| 工程概论        | √                | √   |     |     |     | √   |     |     |     |      |      |      |
| 微生物学        | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 微生物学实验      | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 化工原理        | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 化工原理实验（一）   | √                | √   | √   |     |     |     |     |     | √   | √    |      |      |
| 酶工程         | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 生化反应工程原理    | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 基因工程        | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 基因工程实验      | √                | √   | √   |     |     |     |     |     | √   | √    |      |      |
| 发酵工程        | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 发酵工程实验      | √                | √   | √   |     |     |     |     |     | √   | √    |      |      |
| 生物分离工程      | √                | √   | √   |     |     |     |     |     |     |      |      |      |
| 生物分离工程实验    | √                | √   | √   |     |     |     |     |     | √   | √    |      |      |
| 生物工程工厂设计概论  | √                | √   | √   |     |     |     | √   | √   |     |      | √    |      |
| 生物制药技术      | √                |     |     |     |     |     | √   |     |     |      |      |      |
| 药剂学         | √                |     |     |     |     |     |     |     |     |      |      |      |
| 生物催化与转化     | √                |     |     |     |     |     | √   |     |     |      |      |      |
| 生物炼制        | √                |     |     |     |     |     | √   |     |     |      |      |      |
| 食品生物技术      | √                |     |     |     |     |     |     |     |     |      |      |      |
| 药物化学        | √                |     |     |     |     |     |     |     |     |      |      |      |
| 分子生物学       | √                |     |     |     |     |     |     |     |     |      |      |      |
| 生物信息学       | √                |     |     |     | √   |     |     |     |     |      |      | √    |
| 专业英语与科技论文写作 | √                | √   | √   |     |     |     |     |     |     |      |      | √    |
| 细胞工程        | √                |     |     |     |     |     |     |     |     |      |      |      |

| 课程名称        | 生物工程专业“产业计划”毕业要求 |     |     |     |     |     |     |     |     |      |      |      |
|-------------|------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
|             | (1)              | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| 环境生物技术与安全工程 | √                |     |     |     |     | √   | √   |     |     |      |      |      |
| 合成生物学       | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 免疫学基础       | √                | √   |     |     |     |     |     |     |     |      |      |      |
| 认识实习        | √                | √   |     |     |     | √   |     | √   | √   | √    |      | √    |
| 生产实习        | √                | √   |     |     |     | √   |     | √   | √   | √    |      | √    |
| 毕业实习        | √                | √   |     |     |     | √   |     | √   | √   | √    |      | √    |
| 毕业设计(论文)    | √                | √   | √   | √   | √   | √   |     | √   |     | √    |      | √    |
| 生物工程专业设计    | √                | √   | √   | √   |     |     | √   | √   | √   |      | √    |      |
| 创新创业教育      |                  |     |     |     |     | √   | √   |     | √   |      |      |      |
| 第二课堂        | √                |     |     |     |     | √   | √   |     | √   |      | √    | √    |

十、课程修读进程表  
(四年制本科 2021级 生物工程 专业“产业计划”)

X.Course review Process Map



## 十一、化工与制药类教学环节设置及学分分布表（四年制本科 2021级 化工与制药类）

## XI、Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质       | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                                   | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------|------------------|----------------------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                |                  |                                                                                                    |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平<br>台           | 公共<br>课程<br>平台 | 必修               | 1401010<br>大学英语(一)<br>College English (I)                                                          | 2      | 32     | 32      | 0   | 0   | 0                | 1      |                        |                |
|                  |                |                  | 1501882<br>体育(一)<br>Physical Education(I)                                                          | 1      | 26     | 26      | 0   | 0   | 0                | 1      |                        |                |
|                  |                |                  | 5105017<br>思想道德与法治<br>Ideology and morality and rule of law                                        | 3      | 48     | 42      | 0   | 0   | 6                | 1      |                        |                |
|                  |                |                  | 5106001<br>形势与政策<br>World Affairs and State Policy                                                 | 2      | 64     | 64      | 0   | 0   | 0                | 1-8    |                        |                |
|                  |                |                  | 1401841<br>大学英语(二)<br>College English (II)                                                         | 3      | 48     | 48      | 0   | 0   | 0                | 2      |                        |                |
|                  |                |                  | 1501883<br>体育(二)<br>Physical Education(II)                                                         | 1      | 34     | 34      | 0   | 0   | 0                | 2      |                        |                |
|                  |                |                  | 5103001<br>中国近现代史纲要<br>An Outline of Modern and Contemporary<br>History of China                   | 3      | 48     | 42      | 0   | 0   | 6                | 2      |                        |                |
|                  |                |                  | 1401011<br>大学英语(三)<br>College English (III)                                                        | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                        |                |
|                  |                |                  | 1501884<br>体育(三)<br>Physical Education(III)                                                        | 1      | 34     | 34      | 0   | 0   | 0                | 3      |                        |                |
|                  |                |                  | 5102001<br>马克思主义基本原理<br>Fundamentals of Marxism                                                    | 3      | 48     | 44      | 0   | 0   | 4                | 3      |                        |                |
|                  |                |                  | 1401012<br>大学英语(四)<br>College English (IV)                                                         | 2      | 32     | 32      | 0   | 0   | 0                | 4      |                        |                |
|                  |                |                  | 1501885<br>体育(四)<br>Physical Education(IV)                                                         | 1      | 34     | 34      | 0   | 0   | 0                | 4      |                        |                |
|                  |                |                  | 5101001<br>毛泽东思想与中国特色社会主义理论体系概论<br>Theoretical System of Socialism with<br>Chinese Characteristics | 5      | 80     | 64      | 0   | 0   | 16               | 4      |                        |                |
|                  | 通识<br>教育<br>课程 | 必修               | 1306009<br>大学计算机基础<br>Computer Foundation                                                          | 2      | 32     | 20      | 0   | 12  | 0                | 1      |                        |                |
|                  |                |                  | 2502006<br>大学生心理健康教育<br>Mental Health Education                                                    | 2      | 32     | 24      | 0   | 0   | 8                | 1      |                        |                |
|                  |                |                  | 8001001<br>创业学基础<br>Fundamentals of Entrepreneurship                                               | 1      | 16     | 16      | 0   | 0   | 0                | 1, 2   |                        | 滚动开课           |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质 | 课<br>程<br>编<br>码                                                             | 课<br>程<br>名<br>称                                       | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------|------------------------------------------------------------------------------|--------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |          |                                                                              |                                                        |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 公共<br>课程<br>平台   | 必修       | 2503001                                                                      | 职业生涯规划与就业指导<br>Career Planning and Employment Guidance | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |          | 2504005                                                                      | 军事理论<br>Military Theory                                | 2      | 36     | 36      | 0   | 0   | 0                | 2      |                        |                |
|                  |          | 2501002                                                                      | 公益劳动<br>Community Service                              | 1      | 1(周)   | 0       | 0   | 0   | 1(周)             | 4      |                        | 分散进行           |
|                  |          | 思想政治及新时代素质教育类 2 学分<br>Ideological and Political Education 2 Academic Credits |                                                        |        |        |         |     |     |                  |        |                        |                |
|                  |          | 美育教育类 2 学分<br>Art Education 2 Academic Credits                               |                                                        |        |        |         |     |     |                  |        |                        |                |
|                  |          | 经济管理类 1 学分<br>Economic and Management 1 Academic Credit                      |                                                        |        |        |         |     |     |                  |        |                        |                |
|                  |          | 医学类 1 学分<br>Medicine 1 Academic Credit                                       |                                                        |        |        |         |     |     |                  |        |                        |                |
|                  |          | 人文社科类 1 学分<br>Humanity and Social Science 1 Academic Credit                  |                                                        |        |        |         |     |     |                  |        |                        |                |
|                  |          | 0302609                                                                      | 工程制图B<br>Engineering Drawing B                         | 3      | 48     | 40      | 0   | 8   | 0                | 1      |                        |                |
|                  |          | 0702603                                                                      | 高等数学B(一)<br>Advanced Mathematics B(I)                  | 4      | 64     | 64      | 0   | 0   | 0                | 1      |                        |                |
|                  |          | 2202156                                                                      | 现代化工导论<br>Introduction to Modern Chemical Industry     | 1      | 16     | 16      | 0   | 0   | 0                | 1      |                        |                |
|                  |          | 2204100                                                                      | 生物工程导论<br>Introduction to Bioengineering               | 1      | 16     | 16      | 0   | 0   | 0                | 1      |                        |                |
|                  |          | 2206661                                                                      | 无机化学A(一)<br>Inorganic Chemistry A(I)                   | 2.5    | 40     | 40      | 0   | 0   | 0                | 1      |                        |                |
|                  |          | 2253018                                                                      | 无机化学实验A(一)<br>Experiments in Inorganic Chemistry A(I)  | 1      | 16     | 0       | 16  | 0   | 0                | 1      |                        |                |
|                  |          | 0702604                                                                      | 高等数学B(二)<br>Advanced Mathematics B(II)                 | 5      | 80     | 80      | 0   | 0   | 0                | 2      |                        |                |
| 平台               | 必修       | 0703605                                                                      | 大学物理B(一)<br>College Physics B(I)                       | 2.5    | 40     | 40      | 0   | 0   | 0                | 2      |                        |                |
|                  |          | 2206662                                                                      | 无机化学A(二)<br>Inorganic Chemistry A(II)                  | 1.5    | 24     | 24      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                                                                              |                                                        |        |        |         |     |     |                  |        |                        |                |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质             | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                               | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------------|------------------|--------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                      |                  |                                                                                |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平台               | 大类<br>学科<br>基础<br>课程 | 必修               | 2206681<br>分析化学B<br>Analytical Chemistry B                                     | 2      | 32     | 32      | 0   | 0   | 0                | 2      |                        |                |
|                  |                      |                  | 2253019<br>无机化学实验A(二)<br>Experiments in Inorganic Chemistry A(II)              | 1      | 16     | 0       | 16  | 0   | 0                | 2      |                        |                |
|                  |                      |                  | 2253051<br>分析化学实验B<br>Analytical Chemical Experiment B                         | 1      | 24     | 0       | 24  | 0   | 0                | 2      |                        |                |
|                  |                      | 选修               | 1306010<br>C语言程序设计基础<br>Basics of C Programming Language                       | 2.5    | 40     | 40      | 0   | 0   | 0                | 2      |                        |                |
|                  |                      |                  | 1306011<br>C语言程序设计基础实验<br>Experiments on Basics of C Programming Language      | 1      | 24     | 0       | 0   | 24  | 0                | 2      |                        |                |
|                  |                      |                  | 1908696<br>工程美学导论<br>Introduction to Engineering Aesthetics                    | 1.5    | 24     | 24      | 0   | 0   | 0                | 2      |                        |                |
|                  | 学科<br>基础<br>平台       | 必修               | 0401039<br>电路电子技术<br>Electric Circuit and Electronic Technologies              | 3      | 48     | 48      | 0   | 0   | 0                | 3      | 是                      |                |
|                  |                      |                  | 0450005<br>电路电子技术实验<br>Electric Circuit and Electronic Technologies Experiment | 1      | 16     | 0       | 16  | 0   | 0                | 3      | 是                      |                |
|                  |                      |                  | 0702104<br>线性代数B<br>Linear Algebra B                                           | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                        |                |
|                  |                      |                  | 0703606<br>大学物理B(二)<br>College Physics B(II)                                   | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                        |                |
|                  |                      |                  | 0703607<br>大学物理实验B<br>Experiments of College Physics B                         | 1      | 24     | 0       | 24  | 0   | 0                | 3      | 是                      |                |
|                  |                      |                  | 0701061<br>工程力学C<br>Engineering Mechanics C                                    | 2      | 32     | 32      | 0   | 0   | 0                | 4      | 是                      |                |
|                  |                      |                  | 0702019<br>数学建模<br>Mathematical Modeling                                       | 2      | 32     | 32      | 0   | 0   | 0                | 4      | 是                      |                |
|                  |                      |                  | 0702304<br>概率论与数理统计B<br>Probability and Mathematical Statistics(B)             | 2.5    | 40     | 40      | 0   | 0   | 0                | 4      | 是                      |                |
|                  |                      |                  | 2202040<br>化工仪表与自动化<br>Chemical Engineering Instrument and Automation          | 2      | 32     | 32      | 0   | 0   | 0                | 5      | 是                      |                |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课<br>程<br>性<br>质           | 课<br>程<br>编<br>码                                                                       | 课<br>程<br>名<br>称 | 学<br>分                                                                | 合<br>计 | 课 内 学 时 |        |        | 实<br>践<br>学<br>时 | 学<br>期 | 是<br>否<br>辅<br>修/<br>双<br>学<br>位<br>课<br>程 | 先<br>修<br>课<br>程/备<br>注    |
|------------------|----------------------------|----------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------|--------|---------|--------|--------|------------------|--------|--------------------------------------------|----------------------------|
|                  |                            |                                                                                        |                  |                                                                       |        | 讲<br>课  | 实<br>验 | 上<br>机 |                  |        |                                            |                            |
| 平<br>台           | 学<br>科<br>基<br>础<br>平<br>台 | 生<br>物<br>工<br>程<br>(<br>产<br>业<br>计<br>划<br>)<br>专<br>业<br>学<br>科<br>基<br>础<br>课<br>程 | 2203008          | 分离工程A<br>Separation Engineering A                                     | 2      | 32      | 32     | 0      | 0                | 0      | 6                                          |                            |
|                  |                            |                                                                                        | 0703606          | 大学物理B(二)<br>College Physics B(II)                                     | 2      | 32      | 32     | 0      | 0                | 0      | 3                                          |                            |
|                  |                            |                                                                                        | 0703607          | 大学物理实验B<br>Experiments of College Physics B                           | 1      | 24      | 0      | 24     | 0                | 0      | 3                                          |                            |
|                  |                            |                                                                                        | 2204012          | 生物化学<br>Biochemistry                                                  | 4      | 64      | 64     | 0      | 0                | 0      | 3                                          |                            |
|                  |                            |                                                                                        | 2206675          | 物理化学B(一)<br>Physical Chemistry B(I)                                   | 2      | 32      | 32     | 0      | 0                | 0      | 3                                          | 无机化学<br>A(二), 无机<br>化学A(一) |
|                  |                            |                                                                                        | 2206679          | 有机化学B<br>Organic Chemistry B                                          | 2.5    | 40      | 40     | 0      | 0                | 0      | 3                                          |                            |
|                  |                            |                                                                                        | 2253027          | 有机化学实验B<br>Organic Chemical Experiment B                              | 1      | 24      | 0      | 24     | 0                | 0      | 3                                          |                            |
|                  |                            |                                                                                        | 2253054          | 物理化学实验B(一)<br>Experiments in Physical Chemistry B(I)                  | 1      | 24      | 0      | 24     | 0                | 0      | 3                                          |                            |
|                  |                            |                                                                                        | 2204017          | 微生物学<br>Microbiology                                                  | 3      | 48      | 48     | 0      | 0                | 0      | 4                                          |                            |
|                  |                            |                                                                                        | 2206676          | 物理化学B(二)<br>Physical Chemistry B(II)                                  | 1.5    | 24      | 24     | 0      | 0                | 0      | 4                                          | 无机化学<br>A(二), 无机<br>化学A(一) |
|                  |                            |                                                                                        | 2253026          | 物理化学实验B(二)<br>Experiments in Physical Chemistry B(II)                 | 1      | 16      | 0      | 16     | 0                | 0      | 4                                          |                            |
|                  |                            |                                                                                        | 2203015          | 化工原理<br>Principles of Chemical Engineering                            | 4      | 64      | 64     | 0      | 0                | 0      | 5                                          |                            |
|                  |                            |                                                                                        | 2250008          | 化工原理实验(一)<br>Experiments of Principles of Chemical<br>Engineering (I) | 1      | 24      | 0      | 24     | 0                | 0      | 5                                          |                            |
|                  |                            |                                                                                        | 0401001          | 电工技术<br>Electrotechnics                                               | 2      | 32      | 24     | 8      | 0                | 0      | 3                                          |                            |
|                  |                            |                                                                                        | 0702104          | 线性代数B<br>Linear Algebra B                                             | 2      | 32      | 32     | 0      | 0                | 0      | 3                                          |                            |
|                  |                            |                                                                                        | 0502004          | 管理学原理<br>Principles of Management                                     | 3      | 48      | 48     | 0      | 0                | 0      | 4                                          |                            |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质               | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称 | 学<br>分        | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|------------------------|------------------|------------------|---------------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                        |                  |                  |               |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平<br>台           | 生物<br>工程<br>(产业<br>计划) | 选<br>修           | 0702304          | 概率论与数理统计B     | 2.5    | 40      | 40  | 0   | 0                | 0      | 4                      |                |
|                  |                        |                  | 1601004          | 信息检索与利用       | 1      | 16      | 16  | 0   | 0                | 0      | 4                      |                |
|                  |                        |                  | 2202061          | 实验设计与数据处理     | 2      | 32      | 32  | 0   | 0                | 0      | 4                      |                |
|                  |                        |                  | 2202081          | 计算机辅助设计 (CAD) | 2      | 32      | 24  | 0   | 8                | 0      | 5                      |                |
| 模<br>块           | 专业<br>课程<br>模块         | 必<br>修           | 2206667          | 物理化学A(一)      | 2.5    | 40      | 40  | 0   | 0                | 0      | 3                      |                |
|                  |                        |                  | 2206671          | 有机化学A         | 4      | 64      | 64  | 0   | 0                | 0      | 3                      |                |
|                  |                        |                  | 2253052          | 物理化学实验A(一)    | 1.5    | 32      | 0   | 32  | 0                | 0      | 3                      |                |
|                  |                        |                  | 2253055          | 有机化学实验A       | 1.5    | 36      | 0   | 36  | 0                | 0      | 3                      | 是              |
|                  |                        |                  | 2202162          | 化工热力学B        | 2      | 32      | 32  | 0   | 0                | 0      | 4                      | 是              |
|                  |                        |                  | 2206668          | 物理化学A(二)      | 2      | 32      | 32  | 0   | 0                | 0      | 4                      | 是              |
|                  |                        |                  | 2253053          | 物理化学实验A(二)    | 1      | 24      | 0   | 24  | 0                | 0      | 4                      | 是              |
|                  |                        |                  | 2202164          | 能源化学B         | 2      | 32      | 32  | 0   | 0                | 0      | 5                      | 是              |
|                  |                        |                  | 2202165          | 化工环保与安全技术     | 2      | 32      | 32  | 0   | 0                | 0      | 5                      | 是              |
|                  |                        |                  | 2202171          | 化工设备与材料A      | 3      | 48      | 48  | 0   | 0                | 0      | 5                      | 是              |
|                  |                        |                  | 2203604          | 化工原理(一)       | 3      | 48      | 48  | 0   | 0                | 0      | 5                      | 是              |
|                  |                        |                  | 2250008          | 化工原理实验(一)     | 1      | 24      | 0   | 24  | 0                | 0      | 5                      | 是              |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质                                                             | 课<br>程<br>编<br>码                                      | 课<br>程<br>名<br>称                                                                      | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期  | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|---------|------------------------|----------------|
|                  |                                                                      |                                                       |                                                                                       |        |        | 讲 课     | 实 验 | 上 机 |                  |         |                        |                |
| 模<br>块           | 专业<br>课程<br>模块                                                       | 化学<br>工程<br>与<br>工<br>艺<br>专<br>业<br>必<br>修<br>课<br>程 | 2251008<br>能源化学实验<br>Experiment of Energy Chemistry                                   | 1      | 16     | 0       | 16  | 0   | 0                | 5       | 是                      |                |
|                  |                                                                      |                                                       | 2202166<br>化工专业实验<br>Chemical Specialty Experiments                                   | 2      | 48     | 0       | 48  | 0   | 0                | 5, 6, 7 | 是                      |                |
|                  |                                                                      |                                                       | 2202071<br>化工工程设计与技术经济分析<br>Chemical Engineering Design and Techno-Economics Analysis | 2      | 32     | 32      | 0   | 0   | 0                | 6       | 是                      |                |
|                  |                                                                      |                                                       | 2202167<br>化工过程分析与合成B<br>Chemical process analysis and integration B                  | 2      | 32     | 32      | 0   | 0   | 0                | 6       | 是                      |                |
|                  |                                                                      |                                                       | 2202169<br>化学反应工程实验<br>Experiments of Chemical Reaction Engineering                   | 0.5    | 12     | 0       | 12  | 0   | 0                | 6       | 是                      |                |
|                  |                                                                      |                                                       | 2203005<br>化学反应工程A<br>Chemical Reaction Engineering A                                 | 2.5    | 40     | 40      | 0   | 0   | 0                | 6       | 是                      |                |
|                  |                                                                      |                                                       | 2203605<br>化工原理(二)<br>Principles of Chemical Engineering (II)                         | 3      | 48     | 48      | 0   | 0   | 0                | 6       | 是                      |                |
|                  |                                                                      |                                                       | 2250009<br>化工原理实验(二)<br>Experiments of Principles of Chemical Engineering (II)        | 1      | 24     | 0       | 24  | 0   | 0                | 6       | 是                      |                |
|                  | 生物<br>工程<br>(<br>产<br>业<br>计<br>划<br>)<br>专<br>业<br>必<br>修<br>课<br>程 | 必<br>修                                                | 2204007<br>基因工程<br>Gene Engineering                                                   | 2      | 32     | 32      | 0   | 0   | 0                | 5       |                        | 微生物学           |
|                  |                                                                      |                                                       | 2204040<br>生物化学实验<br>Experiments in Biochemistry                                      | 1      | 24     | 0       | 24  | 0   | 0                | 5       |                        |                |
|                  |                                                                      |                                                       | 2204101<br>生化反应工程原理<br>Principles of Biochemical reaction Engineering                 | 2      | 32     | 32      | 0   | 0   | 0                | 5       |                        | 生物化学           |
|                  |                                                                      |                                                       | 2204103<br>生物分离工程<br>Bioseparation Engineering                                        | 2      | 32     | 32      | 0   | 0   | 0                | 5       |                        | 生物化学, 微生物学     |
|                  |                                                                      |                                                       | 2204108<br>基因工程实验<br>Experiments in Gene Engineering                                  | 1      | 24     | 0       | 24  | 0   | 0                | 5       |                        |                |
|                  |                                                                      |                                                       | 2204109<br>生物分离工程实验<br>Experiments in Bioseparation Engineering                       | 1      | 24     | 0       | 24  | 0   | 0                | 5       |                        |                |
|                  |                                                                      |                                                       | 2252002<br>微生物学实验<br>Microbiological Experiments                                      | 1      | 24     | 0       | 24  | 0   | 0                | 5       |                        |                |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质                                 | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                       | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注                      |
|------------------|------------------------------------------|------------------|----------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|-------------------------------------|
|                  |                                          |                  |                                                                                        |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                                     |
| 模<br>块           | 生物<br>工程<br>(产业<br>计划)<br>专业<br>必修<br>课程 | 2204008          | 酶工程<br>Enzyme Engineering                                                              | 2      | 32     | 32      | 0   | 0   | 0                | 6      |                        | 生物化学                                |
|                  |                                          | 2204071          | 生物工程工厂设计概论<br>Introduction to Bioengineering<br>Factory Design                         | 2      | 32     | 32      | 0   | 0   | 0                | 6      |                        | 化工原理                                |
|                  |                                          | 2204102          | 发酵工程<br>Fermentation Engineering                                                       | 2      | 32     | 32      | 0   | 0   | 0                | 6      |                        | 生化反应<br>工程原理,<br>微生物学               |
|                  |                                          | 2204110          | 发酵工程实验<br>Experiments in Fermentation Engineering                                      | 1      | 24     | 0       | 24  | 0   | 0                | 6      |                        |                                     |
|                  |                                          | 2204006          | 生物工程专业实验<br>Bioengineering Professional Experiments                                    | 2      | 48     | 0       | 48  | 0   | 0                | 7      |                        | 生物化学<br>实验, 微生物<br>学实验              |
|                  |                                          | 2204045          | 生物工程设备<br>Biochemical Engineering Equipments                                           | 2      | 32     | 32      | 0   | 0   | 0                | 7      |                        | 发酵工程,<br>生物分离<br>工程                 |
|                  | 专业<br>课程<br>模块                           | 2202061          | 实验设计与数据处理<br>Experiment Design and Data Processing                                     | 2      | 32     | 32      | 0   | 0   | 0                | 4      |                        |                                     |
|                  |                                          | 2202160          | 药物化学与新药研发<br>Medicinal Chemistry and New Drug<br>Development                           | 2      | 32     | 32      | 0   | 0   | 0                | 4      |                        |                                     |
|                  |                                          | 2206012          | 应用催化<br>Applied Catalyzation                                                           | 2      | 32     | 32      | 0   | 0   | 0                | 4      |                        |                                     |
|                  |                                          | 2202003          | 材料化学<br>Materials Chemistry                                                            | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                        |                                     |
|                  |                                          | 2202018          | 精细有机合成及工艺学<br>Fine Organic Synthesization and<br>Technology                            | 2      | 32     | 32      | 0   | 0   | 0                | 5      | 是                      | 仅限化学<br>工程与工<br>艺专业精<br>细化工方<br>向选修 |
|                  |                                          | 2202019          | 绿色化学<br>Green Chemistry                                                                | 1.5    | 24     | 24      | 0   | 0   | 0                | 5      |                        |                                     |
|                  |                                          | 2202021          | 纳米材料基础<br>Basics of Nanomaterials                                                      | 1.5    | 24     | 24      | 0   | 0   | 0                | 5      |                        |                                     |
|                  |                                          | 2202044          | 专业英语<br>Specialized English                                                            | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                        |                                     |
|                  |                                          | 2202049          | 计算机辅助化工制图与识图<br>Computer Aided Drawing and Drawings<br>Reading in Chemical Engineering | 2      | 32     | 24      | 0   | 8   | 0                | 5      |                        |                                     |
|                  | 专业<br>选修<br>课程                           |                  |                                                                                        |        |        |         |     |     |                  |        |                        |                                     |
|                  |                                          |                  |                                                                                        |        |        |         |     |     |                  |        |                        |                                     |
|                  |                                          |                  |                                                                                        |        |        |         |     |     |                  |        |                        |                                     |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质       | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                     | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注      |
|------------------|----------------|------------------|--------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|---------------------|
|                  |                |                  |                                                                                      |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                     |
| 模块               | 专业<br>课程<br>模块 | 专业<br>选修<br>课程   | 2202157<br>化学制药工艺学<br>Chemical Pharmaceutical Technology                             | 2.5    | 40     | 40      | 0   | 0   | 0                | 5      | 是                      | 仅限化学工程与工艺专业制药方向选修   |
|                  |                |                  | 2202174<br>炭材料科学与工程基础A<br>Science and Engineering Fundamentals of Carbon Materials A | 2.5    | 40     | 40      | 0   | 0   | 0                | 5      | 是                      | 仅限化学工程与工艺专业煤化工方向选修  |
|                  |                |                  | 2206003<br>高分子物理与化学<br>Polymer Physics and Chemistry                                 | 1.5    | 24     | 24      | 0   | 0   | 0                | 5      |                        |                     |
|                  |                |                  | 2251005<br>化学开放实验<br>Opening Chemistry Experiment                                    | 1.5    | 24     | 24      | 0   | 0   | 0                | 5      |                        |                     |
|                  |                |                  | 2202006<br>高分子材料加工工艺学<br>Polymer Materials Processing Technology                     | 2      | 32     | 32      | 0   | 0   | 0                | 6      |                        |                     |
|                  |                |                  | 2202029<br>燃气工程<br>Gas Engineering                                                   | 2      | 32     | 32      | 0   | 0   | 0                | 6      | 是                      | 仅限化学工程与工艺专业煤化工方向选修  |
|                  |                |                  | 2202064<br>化学产品设计基础<br>Chemical Product Design fundamental                           | 1.5    | 24     | 24      | 0   | 0   | 0                | 6      |                        |                     |
|                  |                |                  | 2202075<br>石油化学与石油炼制工程<br>Petrochemical and petroleum refining engineering           | 2      | 32     | 32      | 0   | 0   | 0                | 6      | 是                      | 仅限化学工程与工艺专业精细化工方向选修 |
|                  |                |                  | 2202159<br>药品生产质量管理工程<br>Pharmaceutical Production Quality Management Engineering    | 2      | 32     | 32      | 0   | 0   | 0                | 6      | 是                      | 仅限化学工程与工艺专业制药方向选修   |
|                  |                |                  | 2202161<br>化工过程计算机模拟<br>Computer Simulation in Chemical Process                      | 2      | 32     | 24      | 0   | 8   | 0                | 6      |                        |                     |
|                  |                |                  | 2202610<br>能源化工工艺学<br>Energy Chemical technics                                       | 2      | 32     | 32      | 0   | 0   | 0                | 6      | 是                      | 仅限化学工程与工艺专业煤化工方向选修  |
|                  |                |                  | 2202611<br>化工工艺学<br>Chemical Techniques                                              | 2.5    | 40     | 40      | 0   | 0   | 0                | 6      | 是                      | 仅限化学工程与工艺专业精细化工方向选修 |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质       | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                               | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注    |
|------------------|----------------|------------------|--------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|-------------------|
|                  |                |                  |                                                                                |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                   |
| 模<br>块           | 专业<br>课程<br>模块 | 专业<br>选修<br>课程   | 2203001<br>传递过程基础<br>Fundamentals of Transfer Process                          | 2      | 32     | 32      | 0   | 0   | 0                | 6      |                        |                   |
|                  |                |                  | 2203002<br>化工工程放大原理<br>Scale-up Principles of Chemical Engineering             | 1.5    | 24     | 24      | 0   | 0   | 0                | 6      |                        |                   |
|                  |                |                  | 2250007<br>化工仿真实验<br>Chemical Engineering Simulation Experiments               | 1.5    | 24     | 24      | 0   | 0   | 0                | 6      |                        |                   |
|                  |                |                  | 2202028<br>燃料燃烧<br>Fuel Burning                                                | 2      | 32     | 32      | 0   | 0   | 0                | 7      |                        |                   |
|                  |                |                  | 2202056<br>现代分析与测试技术<br>Modern Analysis and Testing Technologies               | 2      | 32     | 32      | 0   | 0   | 0                | 7      |                        |                   |
|                  |                |                  | 2202158<br>工业药剂学<br>Industrial Pharmaceutics                                   | 2      | 32     | 32      | 0   | 0   | 0                | 7      |                        | 仅限化学工程与工艺专业制药方向选修 |
|                  |                |                  | 2206028<br>应用电化学<br>Applied Electrochemistry                                   | 2      | 32     | 32      | 0   | 0   | 0                | 7      |                        |                   |
|                  |                |                  | 2204004<br>分子生物学<br>Molecular Biology                                          | 2      | 32     | 32      | 0   | 0   | 0                | 4      |                        |                   |
|                  |                |                  | 2204018<br>细胞工程<br>Cell Engineering                                            | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                        |                   |
|                  |                |                  | 2204020<br>药物化学<br>Pharmaceutical Chemistry                                    | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                        |                   |
|                  |                |                  | 2204052<br>免疫学基础<br>Immunology Foundation                                      | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                        |                   |
|                  |                |                  | 2204073<br>波谱分析<br>Spectroscopic Analysis                                      | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                        |                   |
|                  |                |                  | 2204104<br>专业英语与科技论文写作<br>Specialized English and writing of scientific papers | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                        | 主要针对生物工程专业        |
|                  |                |                  | 2204023<br>生物信息学<br>Bioinformatics                                             | 2      | 32     | 32      | 0   | 0   | 0                | 6      |                        |                   |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质       | 课<br>程<br>编<br>码                                  | 课<br>程<br>名<br>称 | 学<br>分                                                            | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------|---------------------------------------------------|------------------|-------------------------------------------------------------------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                |                                                   |                  |                                                                   |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 模<br>块           | 专业<br>课程<br>模块 | 专业<br>选修<br>课程                                    | 2204048          | 生物制药技术<br>Biopharmaceutical Manufacturing Technologies            | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                                                   | 2204105          | 环境生物技术与安全工程<br>Environmental Biotechnology and Safety Engineering | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                                                   | 2204106          | 食品生物技术<br>Food Biotechnology                                      | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                                                   | 2204107          | 合成生物学<br>Synthetic Biology                                        | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                                                   | 2204612          | 生物炼制<br>Bio-refineries                                            | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                                                   | 2204058          | 药剂学<br>Pharmaceutics                                              | 2      | 32      | 32  | 0   | 0                | 0      | 7                      |                |
|                  | 实践教学<br>模块     | 大类<br>实践<br>课程                                    | 2504006          | 军事训练<br>Military Training                                         | 2      | 112     | 0   | 0   | 0                | 112    | 1                      |                |
|                  |                |                                                   | 1701009          | 工程训练C<br>Engineering Training C                                   | 1      | 2(周)    | 0   | 0   | 0                | 2(周)   | 2                      | 是              |
|                  |                | 化学<br>工程<br>与<br>工<br>艺<br>专<br>业<br>实<br>践<br>课程 | 2202052          | 认识实习<br>Introductory Practice Experience                          | 1      | 1(周)    | 0   | 0   | 0                | 1(周)   | 5                      | 是              |
|                  |                |                                                   | 2202083          | 工程实践与创新<br>engineering practice and innovation                    | 2      | 4(周)    | 0   | 0   | 0                | 4(周)   | 6                      | 是              |
|                  |                |                                                   | 2203004          | 化工原理课程设计<br>Course Project in Principles of Chemical Engineering  | 1      | 2(周)    | 0   | 0   | 0                | 2(周)   | 6                      | 是              |
|                  |                |                                                   | 2202043          | 生产实习<br>Production Practice                                       | 2      | 2(周)    | 0   | 0   | 0                | 2(周)   | 7                      |                |
|                  |                |                                                   | 2202097          | 毕业实习<br>Pre-graduation Internship                                 | 1      | 1(周)    | 0   | 0   | 0                | 1(周)   | 8                      | 是              |
|                  |                |                                                   | 2202098          | 毕业设计(论文)<br>Undergraduate Project (Thesis)                        | 8      | 14(周)   | 0   | 0   | 0                | 14(周)  | 8                      | 是              |
|                  |                | 生物<br>工程<br>(产业<br>计划)<br>专业<br>实践<br>课程          | 2204005          | 认识实习<br>Introductory Practice                                     | 2      | 2(周)    | 0   | 0   | 0                | 2(周)   | 5                      |                |
|                  |                |                                                   | 2203004          | 化工原理课程设计<br>Course Project in Principles of Chemical Engineering  | 1      | 2(周)    | 0   | 0   | 0                | 2(周)   | 6                      |                |

## 化工与制药类教学环节设置及学分分布表

(四年制本科 2021级 化工与制药类)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质                   | 课<br>程<br>编<br>码           | 课<br>程<br>名<br>称                                        | 学<br>分                                               | 合<br>计    | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------------------|----------------------------|---------------------------------------------------------|------------------------------------------------------|-----------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                            |                            |                                                         |                                                      |           | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 模<br>块           | 实<br>践<br>教<br>学<br>模<br>块 | 必<br>修                     | 2204031<br>生产实习<br>Production Practice                  | 4                                                    | 4(周)      | 0       | 0   | 0   | 4(周)             | 7      |                        |                |
|                  |                            |                            | 2204096<br>生物工程专业设计<br>Project Design of Bioengineering | 1                                                    | 2(周)      | 0       | 0   | 0   | 2(周)             | 7      |                        |                |
|                  |                            |                            | 2204097<br>毕业实习<br>Pre-graduation Internship            | 2                                                    | 2(周)      | 0       | 0   | 0   | 2(周)             | 8      |                        |                |
|                  |                            |                            | 2204098<br>毕业设计(论文)<br>Undergraduate Project(Thesis)    | 8                                                    | 14<br>(周) | 0       | 0   | 0   | 14<br>(周)        | 8      | 是                      |                |
|                  | 素<br>质<br>拓<br>展<br>模<br>块 | 创<br>新<br>创<br>业<br>教<br>育 | 必<br>修                                                  | 创新创业实践3学分<br>Innovation Practices 3 Academic Credits |           |         |     |     |                  |        |                        |                |
|                  |                            | 第<br>二<br>课<br>堂           | 必<br>修                                                  | 第二课堂3学分<br>Second Classroom 3 Academic Credits       |           |         |     |     |                  |        |                        |                |

